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Education

Program	Institution	% / SGPA	Year
B.Tech in Computer Science and Engineering		8.63	2025-26(pursuing)
Diploma in Information Technology	Government Polytechnic , Kolhapur	95.18%	2022-25

Projects

SignBridge | AI-Powered Sign Language Translator

- React.js, Node.js, Express.js, Python, OpenCV, MediaPipe, scikit-learn, TensorFlow

- **Pipeline:** Developed an end-to-end CV pipeline using **MediaPipe** for 21-point hand landmark extraction and spatial feature engineering.
- **Optimization:** Reduced inference latency by batch-processing frames and optimizing the communication bridge between Python (ML) and Node.js (Backend).
- **Model Training:** Evaluated and deployed Random Forest and Neural Network classifiers for 95%+ accuracy in static and dynamic gesture recognition.
- **Accessibility:** Built a multimodal system providing real-time text-to-speech and visual feedback for low-latency user interaction.
- **Architecture:** Structured the application using modular frontend-backend communication pipelines enabling easier model integration and future scalability.

LLMGuard | AI Watermarking & Detection System

- Python, Flask, React.js, PyTorch, Hugging Face Transformers, NumPy

- **Core Logic:** Implemented a LogitsProcessor using Hugging Face to modify probability distributions during auto-regressive generation.
- **Security:** Designed a deterministic vocabulary partitioning scheme using **SHA256-based hashing** to create "green" and "red" token sets.
- **Statistical Analysis:** Built a detection engine using **Z-score hypothesis testing** based on green token concentration to identify AI-generated text.
- **Optimization:** Conducted robustness stress-tests against varying sampling temperatures and top-p/top-k strategies to measure watermark persistence.
- **Architecture:** Decoupled the ML inference engine from the detection module for modular experimentation with different LLM backends.

Park IT | Smart Parking Allotment System

- Flutter, Firebase (Firestore, Authentication), REST API

- **System Design & Real-Time Management:** Developed a real-time smart parking allocation platform for parking discovery, booking, and slot management using Flutter and Cloud Firestore.
- **Authentication & Role-Based Access:** Implemented Firebase Authentication with dual-role access control for users and administrators.
- **Concurrency & Optimization:** Built a synchronized slot availability engine to prevent double booking and optimize low-latency Firestore updates.
- **Location & UI Engineering:** Designed a responsive Flutter UI with location-based parking search and filtering for improved accessibility and user experience.
- **Performance & UI Engineering:** Optimized asynchronous data retrieval and Firestore query handling for smooth cross-device performance and real-time updates.